BAMBINOS.LIVE • HIRING ASSESSMENT (STAGE 2 — OPERATIONS)

AI Program Manager (Operations) — Take-Home Case

Re-Engineering & Automating the Student Renewal → Class Scheduling Pipeline

Role: AI Program Manager (Operations)	**Timebox:** 48–72 Hours	**Coding Required:** None (AI / No-Code / Sheets / SQL logic)
Location: Bengaluru (In-Office)	**Submission:** PDF + GitHub Repo (Form Link)	**Next Stage:** Live Case Review with Head of Operations

1. Business Context & The Current Renewal Flow

Bambinos.live runs thousands of live small-group online classes every week for children aged 4–16 across India, GCC, and US/UK. A massive driver of our revenue and retention is **Student Renewals** — where existing parents pay to continue their child's classes for the next term/level, often pitched directly by their current teacher.

Today, moving a renewing student from **Payment Completed** to **Classes Scheduled** involves four database tables/sheets and multiple manual handoffs across Teachers, Operations, and Parents:

Step & Table	How It Works Today	Owner
Step 1: invoices	The teacher pitches renewal to the parent. Once the parent pays via payment link, an entry is created in the **invoices** table.	Parent / Payment Gateway
Step 2: faculty_form_unapproved	After payment, the teacher must manually fill out a Renewal Form pasting the invoice link, student details, and payment info. This creates a row in **faculty_form_unapproved**.	Teacher (Faculty)
Step 3: booking_requests	The Ops team manually checks each row in *faculty_form_unapproved*, verifies the invoice link/payment in *invoices*, and approves it — creating a row in **booking_requests**.	Operations Team
Step 4: class_schedule_bookings	From *booking_requests*, either the parent or the Ops team selects the batch/days/time to continue with, which finally creates entries in **class_schedule_bookings**.	Parent / Operations Team

Your Challenge: On paper, this 4-step chain looks simple. In reality, at hundreds of renewals per week, parents who have *already paid* frequently experience interrupted classes, delayed scheduling, and frustration while Ops spends hours doing repetitive manual verification and chasing teachers/parents.

2. Required Deliverables

Deliverable A: Root-Cause Problem Definition & Leakage Audit

- **Spot the Hidden Failure Modes:** Analyze the 4-table flow above. Where exactly do students get stuck, dropped, or delayed between *invoices* → *faculty_form_unapproved* → *booking_requests* → *class_schedule_bookings*? (Think about human behavior, data mismatches, typos, weekend delays, same-batch vs. batch-change renewals, partial payments, etc.).
- **Quantify the Risk:** Define the key operational metrics/SLAs that are currently broken or invisible (e.g., 'Orphaned Invoice Rate', 'Payment-to-Schedule Turnaround Time').

Deliverable B: AI & No-Code Re-Engineered Workflow (The 'Zero-Touch' Ideal)

- **Eliminate & Automate:** Redesign the workflow using AI and no-code automation (n8n, Make, Zapier, LLM agents, Google Sheets/Metabase, WhatsApp API). Which manual handoffs can be completely eliminated for 'happy-path' renewals (e.g., a child continuing in the same batch with the same teacher)?
- **Smart Exception Handling:** When data doesn't match (e.g., parent paid using a different phone number/email than the student record, or teacher submitted the wrong invoice amount), how does your AI/automation triage the discrepancy and alert the exact right person?

Deliverable C: Working Prototype (Reconciliation & Auto-Approval Engine)

- **Use the Provided 4-Table Dataset (*./ops/data/*):** Access the official assessment repository at https://github.com/bambinoslive/AI-PM-hiring-take-home-assignments and use the 5 CSV files provided in *./ops/data/* (*1_invoices.csv, 2_faculty_form_unapproved.csv, 3_booking_requests.csv, 4_class_schedule_bookings.csv, and reference_active_batches_and_students.csv*).
- **Build the AI / Automation Prototype:** Build a working prompt, script, Custom GPT, or n8n/Make/Sheet automation that ingests these CSVs and outputs: (1) Auto-approved entries ready for *class_schedule_bookings*, (2) Flagged exceptions with exact reason codes, and (3) Auto-drafted WhatsApp/Slack follow-up messages for the teacher, parent, or Ops agent.

Deliverable D: Live Ops Dashboard Design, SOP & 30-60-90 Day Plan

- **Ops Head Dashboard:** Sketch or build the daily tracker/dashboard the Head of Operations sees every morning (tracking stuck invoices >12h, unapproved forms, unscheduled bookings, and hours saved).
- **Change Management:** How will you roll this out across 200+ teachers and a 10-person Ops team without breaking live class schedules during transition?

3. Evaluation Rubric

Dimension	What We Look For	Weight
1. Problem Discovery & Edge Cases	Uncovering real-world failure modes across the 4 tables (orphaned invoices, teacher delays, ID mismatches, same-batch fast-tracking).	**25%**
2. Process Re-Engineering	Questioning why manual steps exist instead of blindly automating bad steps; designing a fast-track for clean renewals.	**25%**
3. AI & No-Code Execution	Working Sheet/n8n/Make/AI prototype that reconciles messy records, flags exceptions accurately, and drafts actionable alerts.	**25%**
4. Data, SLAs & Ownership	Clear SLA definitions, exception dashboard design, realistic time/cost savings, and practical rollout SOPs for non-tech teams.	**25%**

4. Submission Format & Guidelines

Submission Checklist & Official Links:
- **1. Case Report / Deck in PDF Format (COMPULSORY):** Covering Deliverables A, B, and D (problem audit, redesigned workflow, SOP, dashboard design, and 30-60-90 rollout).
- **2. GitHub Repository Link (COMPULSORY):** Public or shared repo containing your prompts (*.md*), README documentation, simulated 4-table datasets (CSV/JSON), reconciliation results, scripts, or workflow exports.
- **3. Loom Video Walkthrough (OPTIONAL):** A short 2–3 minute screen recording walking through your reconciliation prototype and exception handling.
- **4. Official Asset Repository:** https://github.com/bambinoslive/AI-PM-hiring-take-home-assignments
- **5. Submit Your Assignment Here (Google Form):** https://forms.gle/gu5gzySRHTSd5Uou9